

Social Media & Political Polarization

POLI 209: Analyzing Public Opinion

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Introduction

In the realm of contemporary politics, the influence of social media on shaping individuals' political ideologies has become a subject of considerable interest and concern. Among the demographic most impacted by this phenomenon are college students, whose years spent in college often coincide with the beginning of their political consciousness. Thus, investigating how the use of social media influences college students' political actions is of great importance in understanding the dynamics of contemporary politics.

College students, immersed in the digital age, spend hours scrolling through their smartphones, consuming a plethora of content that inundates their feeds. Yet, amid this deluge of information, lies a critical concern: how much of this content fosters diverse perspectives and critical thinking, and how much simply reinforces existing beliefs? The answer to this question holds significant implications, not only for individual political behaviors but also for broader societal dynamics.

In this context, investigating how social media influences college students' political actions is of paramount importance. It is a juncture where minds are malleable, and exposure to diverse viewpoints can either foster open-mindedness or entrench ideological polarization. The college environment, characterized by its diversity and potential for intellectual growth, offers a unique opportunity to explore the impact of social media on political attitudes and behaviors.

Therefore, this paper seeks to delve into the intricate relationship between social media usage and the political ideologies of college students. By examining the extent to which social media platforms contribute to the polarization of political ideologies among this demographic, we aim to shed light on the broader implications for democratic engagement and discourse in contemporary society. Through this exploration, we endeavor to not only understand how social media shapes political ideologies but also to underscore the importance of promoting critical engagement and inclusivity in the digital age. In doing so, we hope to contribute to a deeper understanding of the evolving dynamics of contemporary politics and the role of social media within it.

Theory

The main dependent variable underlying our research topic is the polarization of political ideologies among college students. Political polarization refers to the extent to which individuals' political attitudes, beliefs, and behaviors are divided along ideological lines. In this context, polarization manifests as the divergence of opinions on key political issues and the reinforcement of partisan identities. Understanding the factors that contribute to this polarization is crucial for comprehending contemporary political dynamics.

The variation in political ideologies among college students can be explained by a multitude of factors, including socio-cultural influences, personal experiences, and exposure to political information. However, our study focuses specifically on the role of social media usage in shaping these ideologies. This will also be an easy way of gathering data, as our target group is college students. According to a Pew study in 2023, 60+% of respondents in the college demographic said they prefer to get their news from social media (News Platform Fact Sheet, 2023). Previous studies regarding this topic used a wider demographic range than we will be able to access. In Echo Chambers and Polarization researchers found that even traditional sources like news and print could polarize individuals. For instance, they found that those who get their news from CNN are 57% more likely to be liberally polarized. On the flip side, those who watch Fox News are 20% less likely to be liberally polarized (Hawdon et. al., 2020). Social media platforms serve as virtual ecosystems where individuals are exposed to a wide range of political content, including news articles, opinion pieces, and user-generated discussions. The insights gained from "Big Social Data" allow for the creation of detailed profiles for social users, delineating their behaviors and psychological traits and modeling their perspectives on events and political decisions with precision (Belcastro et al., 2022).

The algorithms employed by these platforms further amplify this exposure by tailoring content to users' preferences and behaviors. It makes sense that the duration and frequency of exposure to diverse political viewpoints on social media would cause variation in the polarization of college students' ideologies. According to Wang et al., (2020), and their central empirical exercise, "the range of political viewpoints that individuals are willing to consider from social media is a key determinant in the formation of polarized

networks and the emergence of echo chambers. . . ” Prolonged exposure to ideologically homogeneous content is likely to reinforce existing political inclinations and deepen partisan divides. Conversely, exposure to diverse perspectives has the potential to challenge individuals’ preconceived notions and foster ideological openness. Therefore, the nature of social media usage—whether it encourages echo chambers or facilitates exposure to diverse viewpoints—plays a critical role in shaping the ideological landscape of college students.

Research Question

Hence, our research question— “How does the frequency of exposure to diverse political viewpoints on social media impact the polarization of college students’ ideologies?”—explores the intersection between digital technology and the democratic process.

Design

In our survey design, we meticulously crafted questions to delve into the intricate dynamics of social media engagement and its impact on the political ideologies of college students. Our approach included a series of questions aimed at capturing various facets of respondents’ interactions with political content on social media platforms. Employing an observational approach, our survey aimed to capture real-world behaviors and attitudes without intervention. Administered anonymously via Qualtrics, the survey was designed for brevity, ensuring minimal participant burden while maximizing response rates. Delving into topics such as seeking behavior for political content, changes in political beliefs due to social media interactions, sharing behaviors, credibility verification practices, exposure to political content aligned and conflicting with pre-existing beliefs, and perceptions of political polarization on different social media platforms, our questions were meticulously crafted to elicit pertinent insights. For instance, we included questions regarding the frequency of encountering political content aligned with their existing beliefs (align) and content that

challenged their beliefs (conflict). These questions allowed us to gauge the extent to which respondents were exposed to diverse perspectives and the potential for ideological reinforcement or challenge within their online environments. Additionally, we incorporated questions designed to probe respondents' perceptions of whether their social media feeds accurately reflected viewpoints consistent with their own political beliefs (reflect). This line of inquiry provided insights into the phenomenon of echo chambers and the degree of ideological homogeneity within individuals' online networks.

Furthermore, our survey included questions aimed at assessing whether respondents had experienced any changes in their political beliefs or opinions as a result of their interactions or discussions on social media platforms (change in opinion). By exploring the dynamic nature of political attitudes and the potential role of social media in shaping them, we sought to uncover the mechanisms underlying ideological polarization among college students. Understanding how exposure to diverse political viewpoints on social media may influence individuals' beliefs and attitudes over time is essential for comprehending the evolving landscape of contemporary politics. Additionally, we incorporated demographic questions to capture a comprehensive profile of our sample, including factors such as age, gender, year of study, party affiliation, and political ideology. This demographic information provided valuable context for analyzing the relationship between social media usage and ideological polarization among college students, allowing for potential moderating effects to be explored.

Moreover, our survey design was informed by established theoretical frameworks and empirical research on social media's influence on political attitudes and behaviors. By drawing on insights from previous studies, we ensured that our survey questions were theoretically grounded and empirically validated, enhancing the robustness and validity of our findings. Statistical analyses, including regression models and t-tests, were employed to rigorously test hypotheses derived from our framework, providing quantitative evidence to support our conclusions. Through this comprehensive approach, we aimed to contribute nuanced insights to the ongoing discourse on the complex interplay between social media usage and political ideologies among college students, shedding light on the mechanisms driving ideological polarization in the digital age.

With the distribution of our survey came certain limitations. We disseminated the survey primarily among college students within our network, aiming to capture responses from individuals within this specific

demographic. While this approach facilitated targeted data collection, it also introduced a potential bias due to the homogeneous nature of our sample. Most respondents were affiliated with the same educational institution, thus limiting the diversity of perspectives represented in our data. Furthermore, the demographic composition of our sample may not be reflective of the broader population of college students, particularly considering the political orientation of the institution where the survey was distributed, which tends to lean towards liberalism. As we proceed with data analysis and interpretation, it will be essential to acknowledge these limitations and contextualize our findings within the confines of our survey sample, recognizing the potential implications for the generalizability of our results.

Data & Results

Table 1: Modes of Variables

Variable	Mode
Year	3rd year
Instagram	Moderately Liberal
Snapchat	Moderate
Facebook	Moderate
TikTok	Moderately Liberal
Reddit	Moderate
Twitter	Moderate
Align	Frequently
Conflict	Occasionally
Changes	Yes
Reflect	Yes

Among the 114 responses garnered from our survey, the prevailing mode revealed that a significant portion of respondents identified themselves as Juniors in college. In our survey, respondents were asked to specify the media types they utilize for accessing political news. Results unveiled a clear trend, with social media platforms emerging as the preferred choice among respondents, encompassing 79% of those surveyed. This emphasizes the pivotal role social media, particularly platforms like Instagram and Facebook, plays in shaping how college students engage with political information. Following closely behind the dominance of social media, online news websites showcased significant influence, with 69% of respondents identifying them as their primary source for political news. This observation underscores the transformative impact of the digital age on traditional news consumption patterns, highlighting the ascendancy of online platforms in disseminating news and information.

In our findings, word-of-mouth, denoting exchanges within personal networks such as friends, family, and coworkers, emerged as a notable source, selected by 56% of respondents. Conversely, more traditional media outlets like television news channels and print media appeared to garner lower reliance rates, with only 24% and 10% of respondents, respectively, indicating them as primary sources for political news. Similarly, podcasts and radio exhibited comparatively lower levels of reliance, with 19% and 3% of respondents, respectively, identifying them as their primary sources for political news.

Consequently, we conducted an analysis to determine the perceived political polarization associated with various social media platforms among our respondents. The “SMPolarization” variable in our dataset captured respondents’ perceptions, with categories ranging from “Extremely Liberal” to “Extremely Conservative”. Using Rstudio, we processed the dataset to calculate the mode (most frequent response) for each social media platform. The code aggregated the responses for each platform and determined the mode, providing insights into which platforms were perceived as the most liberal, moderate, and conservative.

Our analysis unveiled distinct perceptions of political polarization linked with different social media platforms. Instagram and TikTok were perceived as moderately liberal, while Snapchat, Facebook, Reddit, and Twitter were all viewed as moderate in terms of political leanings. These findings shed light on how users perceive the political landscape of popular social media platforms.

In our data analysis, we utilized R programming to determine the mode for both the “Align” and “Conflict” variables in our polarization dataset. The “Align” variable represents the frequency of encountering political content aligned with respondents’ beliefs on social media, while the “Conflict” variable indicates the frequency of encountering conflicting political content. By employing the mode function, we identified the most common response for each variable.

The results revealed that the mode for “Align” was “Frequently,” indicating that respondents frequently encountered politically aligned content on social media platforms. Conversely, the mode for “Conflict” was “Occasionally,” suggesting that encounters with conflicting political content occurred less frequently compared to aligned content.

Selecting the mode as our statistical measure allowed us to capture the most common response among respondents, providing a succinct summary of their experiences with politically aligned and conflicting content on social media. Moreover, these findings align with our theoretical framework, which posits that social media algorithms often curate content that reinforces users’ existing beliefs, leading to frequent encounters with aligned content and less frequent encounters with conflicting content.

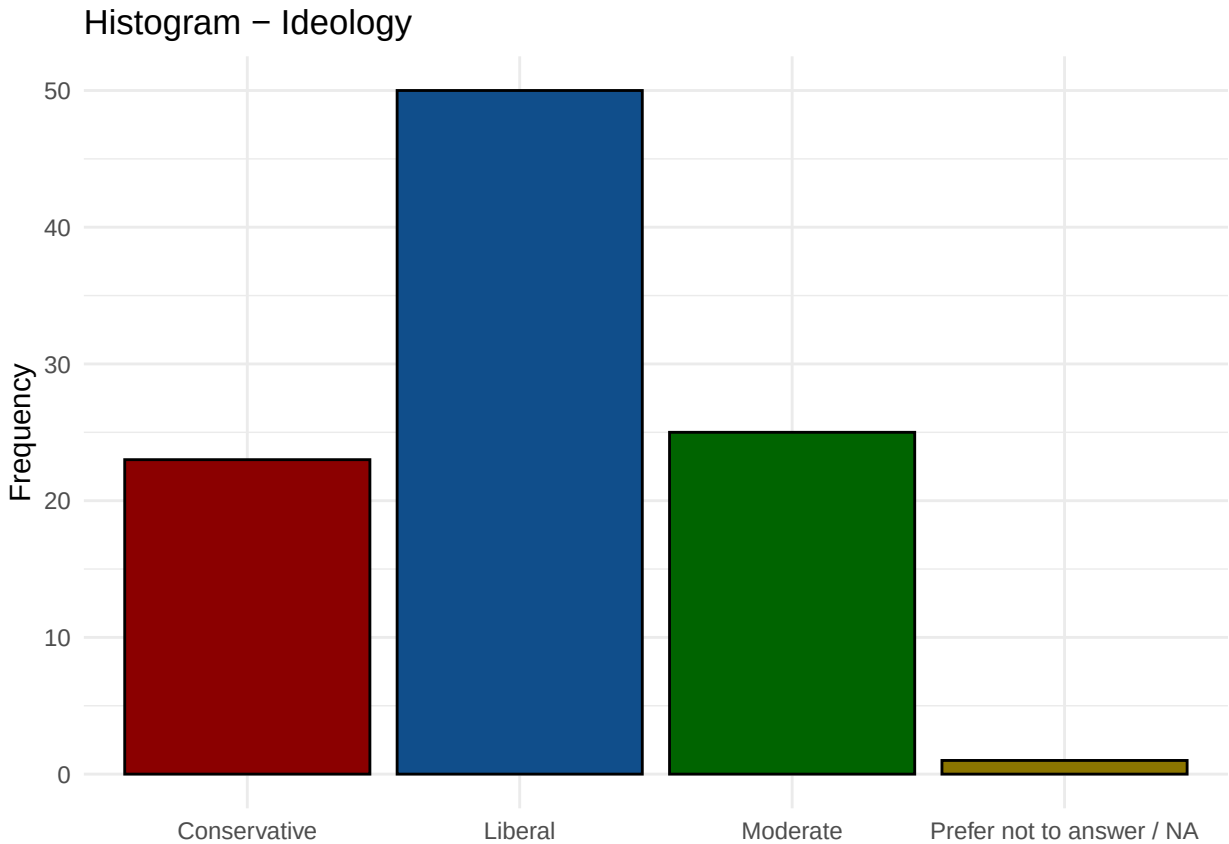
In our analysis, we calculated the mode for the responses in the “Reflect” column of our polarization dataset. The mode represents the most frequently occurring response among the options provided, offering insights into the predominant perception regarding whether social media feeds reflect respondents’ political beliefs. Upon executing the code, the output revealed that the mode for the “Reflect” column was “Yes,” indicating that a significant proportion of respondents perceived that their social media feeds directly reflect viewpoints consistent with their own political beliefs. By choosing the mode as our statistic, we aim to capture the central tendency of responses to the Reflect question, providing a concise summary of how respondents perceive the alignment of their social media feeds with their political beliefs. This method helps simplify complex data into a single, understandable metric, making it easier to identify patterns and trends in how respondents perceive the political content of their social media feeds.

We opted to calculate the mode as our statistic of choice for several reasons. Firstly, the Reflect question is categorical, with response options limited to “Yes,” “No,” and “Unsure / N/A.” As such, the

mode provides a clear representation of the most common response category among respondents, offering a straightforward interpretation of the prevailing sentiment. Additionally, the mode is robust to outliers and does not require assumptions about the distribution of responses, making it suitable for analyzing categorical data.

We tested the “Changes” variable, which queried respondents about any observed alterations in their political beliefs stemming from interactions or discussions on social media. Employing the mode statistical measure, we determined the most frequent response among participants. The mode for “Changes” emerged as “Yes,” indicating that a notable portion of respondents acknowledged experiencing changes in their political beliefs due to their social media engagements.

This finding underscores the profound influence of social media platforms on individuals’ political ideologies. It suggests that these online spaces serve as influential arenas where users engage in discussions, encounter diverse viewpoints, and potentially undergo shifts in their political perspectives.



We began by examining the distribution of respondents’ ideologies using a histogram for the Ideology

variable in our dataset. It became evident that the data exhibited a notable skew, with a predominance of liberal responses. To address this skewness and enhance interpretability, we opted to transform the variable into a binary format. This strategic decision not only simplified the dataset but also facilitated more effective modeling, particularly when conducting our linear regression analysis.

The conversion process involved creating two distinct groups: IdeologyD and IdeologyR. For IdeologyD, respondents were assigned a value of 1 if they identified with either “Somewhat Liberal” or “Extremely Liberal,” and 0 otherwise. Conversely, for IdeologyR, a value of 1 was assigned if respondents identified with either “Somewhat Conservative” or “Extremely Conservative,” and 0 otherwise. This binary categorization enabled us to mitigate bias in subsequent statistical analyses by ensuring more balanced representation across ideological categories.

Subsequently, we conducted two linear regression models to explore the relationship between ideology and changes in political beliefs influenced by social media interactions. In the first model, IdeologyD served as the dependent variable, with Changes (indicating alterations in political beliefs) as the independent variable. In the second model, we conducted an additional regression model, this time focusing on IdeologyR and Changes. This model aimed to explore similar dynamics observed within the conservative population, shedding light on how changes in political beliefs among conservatives were influenced by social media interactions.

Align, serving as an indicator of alignment with political content on social media, was integrated as a control variable to evaluate its influence on ideology alongside changes in political beliefs. The decision to incorporate this variable into our model stemmed from an ANOVA test, yielding a significant p-value of 0.02729, falling below the conventional significance threshold of 0.05. This underscored the Align variable’s statistical significance in predicting the average mean Ideology. Through this analytical approach, we aimed to delve into the intricate relationship between ideology and respondents’ exposure to politically aligned content, as well as the evolution of their political beliefs stemming from interactions on social media platforms.

In examining the model with IdeologyD as the dependent variable, several notable insights emerged. The intercept coefficient, with an estimate of 0.6478 and a standard error of 0.1699, suggests a baseline level

Table 2: Summary Statistics for IdeologyD

	Model 1			
	Estimate	Std. Error	t value	Pr($t >t_{\alpha}$)
(Intercept)	0.6477633	0.1698864	3.8129196	0.0002497
ChangesUnsure / N/A	-0.0789048	0.1744095	-0.4524109	0.6520497
ChangesYes	0.1360354	0.1071806	1.2692166	0.2076004
AlignFrequently	-0.1306154	0.1652836	-0.7902503	0.4314361
AlignOccasionally	-0.4051202	0.1884950	-2.1492356	0.0342694
AlignRarely	-0.6668068	0.3189457	-2.0906596	0.0393468
AlignUnsure / N/A	-0.6763286	0.3779375	-1.7895251	0.0768569

Table 3: Summary Statistics for IdeologyR

	Model 2			
	Estimate	Std. Error	t value	Pr($t >t_{\alpha}$)
(Intercept)	0.3440587	0.1524056	2.2575198	0.0263676
ChangesUnsure / N/A	-0.0248240	0.1564633	-0.1586569	0.8742908
ChangesYes	-0.0781877	0.0961520	-0.8131679	0.4182443
AlignFrequently	-0.1060915	0.1482764	-0.7154983	0.4761331
AlignOccasionally	-0.0062192	0.1690994	-0.0367785	0.9707422
AlignRarely	0.0236119	0.2861271	0.0825225	0.9344125
AlignUnsure / N/A	0.2074472	0.3390488	0.6118505	0.5421623

of ideological alignment, indicating that, on average, respondents identified with liberal ideologies. Similarly, in the regression model with IdeologyR as the dependent variable, our analysis provided insights into the relationship between changes in political beliefs, social media exposure, and conservative political ideology. The intercept coefficient indicates a baseline level of conservative ideological alignment among respondents, with an estimate of 0.3441 and a standard error of 0.1524. However, the coefficients for the “Changes” and “Align” variables were not statistically significant.

Table 4: ANOVA - IdeologyD Model with Align vs without Align

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
95	23.93098	NA	NA	NA	NA
91	21.24836	4	2.682621	2.872204	0.0272877

In the model with IdeologyD as the dependent variable, the p-value for the F-statistic is 0.03999, indicating that the model is statistically significant at the conventional alpha level of 0.05. This suggests

that at least one of the predictor variables (Changes or Align) has a statistically significant relationship with IdeologyD.

However, in the model with IdeologyR as the dependent variable, the p-value for the F-statistic is 0.8469, which is notably higher than 0.05. This suggests that the model as a whole is not statistically significant, and none of the predictor variables (Changes or Align) have a statistically significant relationship with IdeologyR. This lack of significance may stem from the heavily skewed nature of our dataset, where conservative responses were notably underrepresented, thereby limiting the model's ability to discern meaningful relationships.

Discussion & Conclusion

Our study aimed to investigate the intricate relationship between social media usage and the political ideologies of college students, with a particular focus on ideological polarization. Through a comprehensive survey and rigorous statistical analysis, we sought to uncover the mechanisms by which social media platforms influence individuals' political beliefs and behaviors.

Firstly, our analysis revealed a clear preference among respondents for social media platforms as primary sources of political news and information. This underscores the significant role that social media plays in shaping how college students access and consume political content, highlighting the transformative impact of digital technology on traditional news consumption patterns. Moreover, our analysis of perceived political polarization associated with different social media platforms shed light on users' perceptions of the ideological landscape of these platforms. Interestingly, while platforms like Instagram and TikTok were perceived as moderately liberal, others such as Facebook and Twitter were viewed as more politically moderate.

One of the key findings of our study pertains to the influence of algorithms employed by social media platforms in shaping users' online experiences, particularly in the realm of political discourse. These

algorithms play a crucial role in curating users' news feeds and determining the content they are exposed to, often based on their past behavior, preferences, and interactions. As a result, users are more likely to encounter content that aligns with their pre-existing beliefs and preferences, creating what has been termed as "filter bubbles" or "echo chambers."

The prevalence of algorithm-driven content curation has significant implications for political discourse, particularly among college students who are increasingly reliant on social media platforms for news and information. Our study found that a significant portion of respondents perceived their social media feeds as reflecting viewpoints consistent with their own political beliefs, indicating the presence of echo chambers within online networks. While algorithms may enhance user experience by providing personalized content tailored to individual preferences, they also have the potential to exacerbate ideological polarization and contribute to the spread of misinformation. By reinforcing users' existing beliefs and limiting exposure to diverse perspectives, these algorithms can create environments where opposing viewpoints are marginalized or excluded altogether.

Furthermore, our regression analyses provide insights into the relationship between social media exposure, changes in political beliefs, and ideological alignment among college students. While our findings did not uncover significant effects of social media exposure and changes in political beliefs on ideological alignment, they contribute to the ongoing discourse on the influence of social media on political attitudes and beliefs. The lack of statistical significance in the model with IdeologyR as the dependent variable may be attributed to the heavily skewed nature of our dataset, where conservative responses were underrepresented. This limitation highlights the need for future research to explore the impact of social media on conservative political ideologies more comprehensively.

In conclusion, our study offers valuable insights into the complex interplay between social media usage and political ideologies among college students. While our findings did not reveal significant effects of social media exposure on ideological alignment, they underscore the importance of exploring the multifaceted relationship between social media usage and political ideologies.

Ultimately, our study contributes to a deeper understanding of the role of social media in shaping

political attitudes and behaviors, highlighting the need for continued research and dialogue on this pressing issue. By leveraging the insights gained from our study, we can work towards fostering a more informed, engaged, and inclusive political discourse in the digital age.

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